Bihar Engineering University, Patna

B.Tech 7th Semester Examination, 2024

Course: B. Tech
Code : 103701

Subject: Power System Protection

Time: 03 Hours
Full Marks: 70

Instructions: -

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. **1** is compulsory.*

Q.1 Choose the correct answer of the following (any seven question only): **[2 x 7 = 14]**

(a) Which of the following is the primary function of a power system protection scheme?
(i)To regulate voltage levels in the power system
(ii) To detect and isolate faults to prevent system damage
(iii)To enhance power generation efficiency
(iv)To reduce transmission line losses

(b) In a circuit breaker, what is the primary role of the arc-quenching medium?
(i) To provide insulation between conductors
(ii) To detect fault currents
(iii) To extinguish the arc formed during switching operations
(iv) To regulate system voltage

(c) In symmetrical components, which sequence component represents the normal operating condition of a balanced three-phase system?
(i) Zero-sequence component
(ii) Positive-sequence component
(iii) Negative-sequence component
(iv) All of the above

(d) What is the primary function of an overcurrent relay in a power system?
(i) To detect and isolate faults based on voltage variations
(ii) To detect and trip the circuit when current exceeds a preset value
(iii) To regulate frequency in the power system
(iv) To enhance power factor

(e) Distance relays operate based on which of the following parameters?
(i) Voltage only
(ii) Current only
(iii) Impedance between the relay location and fault point
(iv) Frequency variation

(f) Which protection scheme is commonly used to protect transformers from internal faults?
(i) Distance protection
(ii) Overcurrent protection
(iii) Differential protection
(iv) Directional protection

(g) What is the primary advantage of computer-aided protection in power systems?
(i) Reduced power consumption
(ii) Faster and more accurate fault detection
(iii) Elimination of the need for circuit breakers
(iv) Reduction in transformer size

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(h) Which method is commonly used for testing protective relays to ensure proper operation?
(i) Load flow analysis
(ii) Transient stability analysis
(iii) Secondary injection testing
(iv) Harmonic analysis

(i) What is the primary function of an under-frequency relay in power system protection?
(i) To detect and clear faults in transmission lines
(ii) To disconnect loads when frequency drops below a critical level
(iii) To regulate system voltage
(iv) To detect reverse power flow

(j) Phasor Measurement Units (PMUs) rely on which technology for synchronization?
(i) Internet Protocol (IP)
(ii) Global Positioning System (GPS)
(iii) Frequency Modulation (FM)
(iv) Pulse Width Modulation (PWM)

Q.2 (a) Define power system protection. [7]
(b) How does an air circuit breaker differ from an SF6 circuit breaker? [7]

Q.3 (a) Analyze the coordination of overcurrent relays in a system with multiple protection zones. [7]
(b) Describe the difference between symmetrical and unsymmetrical faults in a power system. [7]

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Q.4 (a) Define directional protection and its role in power systems [7]
(b) Discuss the operating principle of differential protection in generators. [7]

Q.5 (a) What is the purpose of using the Discrete Fourier Transform (DFT) in power system protection? [7]
(b) Explain the role of computer-aided protection systems in modern power systems. [7]

Q.6 (a) Describe the significance of transient simulations in analysing power system faults [4]
(b) A three-phase transformer having a line voltage ratio of 400 V/33,000 V is connected in star-delta. The CTs on the line 400 V side have a current ratio of 1000/5. What must be the ratio of CTs on the 33,000 V side. [10]

Q.7 What is the role of Phasor Measurement Units (PMUs) in WAMS? Write Application of WAMS for improving protection systems. [14]

Q.8 Name the different types of relay testing methods used in power system protection and discuss the importance of relay testing in ensuring the reliability of protection schemes. [14]

Q.9 Write short notes on ***any two*** of the following: - [7 x 2 = 14]
(a) Instrument transformers
(b) Differential protection
(c) Computer-aided protection
(d) SF6 circuit breaker

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